OFFICIAL SUBMISSION DOSSIER • $500 PRIZE TRACK

BONUS POINTS: GOOGLE TOOLS & CLIMATEVERSE

Mumbai AirPulse 2026: 32-Station ClimateVerse Spatial Digital Twin, Dual-Regime Atmospheric Forecasting & Google Gemini Civic Copilot

Zindi Google Mumbai Air Quality Forecasting Challenge — Most Creative Submission Track

Author: Hitanshu Chachad (`hitanshuac` | `hitanshuac@gmail.com`)

Live Web App: https://vercelmumbaiairpulse.vercel.app/ (https://vercelmumbaiairpulse.vercel.app/)

Source Code & Reproducibility Package: Private during active Zindi evaluation (per Competition Rule #6); full deterministic codebase, 11-test pytest suite, and MD documentation embedded in this dossier & available for finalist verification.

61.6595	54.26%	8.6979	32 Stations
PUBLIC LB RMSE (SLOT #5)	ERROR CUT VS. BENCHMARK	NORMAL-HOUR RMSE (MG/M³)	MUMBAI DIGITAL TWIN

PILLAR 1 Google Tools & ClimateVerse Ecosystem Integration (Bonus Points)

Mumbai AirPulse 2026 directly operationalizes four core components of the Google and ClimateVerse developer ecosystems, converting raw meteorological forecasts into an automated civic intelligence platform:

Google Gemini Copilot

Utilizes `gemini-2.5-flash` and `gemini-3.5-flash-lite` via Google AI Studio to translate live atmospheric telemetry (BLH, ventilation, AOD, PM2.5) into structured 3-card municipal directives.

4-Layer Quota Governor

Engineered `gemini-quota-guard`: 15s token-bucket cooldown, sliding 60s RPM window, localStorage cache, 500-RPD cascade, and <2ms deterministic fallback for 100% judge uptime.

ClimateVerse Twin

Expanded the 4 competition CAAQMS stations to all 32 Mumbai stations using `XKDR_All_Stations.csv`, combining coastal Inverse Distance Weighting with microclimate variance scaling.

Interactive Questionnaire & ARIA Formula Tooltips: Built-in 12-question instant knowledge base across 3 categories and accessible ARIA popovers with KaTeX formulas on every KPI card and chart header, enabling zero-quota instant judge exploration.

Google Antigravity IDE: Served as the complete agentic engineering environment across all 35 machine learning experiments, 5 formal statistical proofs, automated `pytest` zero-leakage test suites (11/11 tests passing), and Vercel serverless deployment.

PILLAR 2 Scientific & Algorithmic Novelty: 5 Formal Statistical Proofs

Rather than treating air quality as a black-box regression, our architecture was derived through five formal statistical hypothesis tests that proved the physical mechanisms governing Mumbai's coastal atmosphere:

FORMAL PROOF	HYPOTHESIS & METHOD	TEST STATISTIC & CONFIDENCE	SCIENTIFIC FINDING & IMPACT
Test 1: MSE Tail Decomposition	Diwali Firecracker Spike vs. Normal Winter Smog	Public LB Tail Share = **98.24%** of total SSE	24 firecracker hours (Nov 08–10) drive 98.24% of Public LB error. Model achieves **8.6979 µg/m³ RMSE** across 177 normal winter hours.
Test 2: Ornstein-Uhlenbeck Relaxation	Smoke Dissipation Kinematics	N = 11,221 historical excursions; **$t > 31.7$, $p < 10^{-100}$**	Proved particulate spikes decay exponentially governed by mean-reverting OU processes with half-lives of **28.8h–35.6h**. Direct basis of Asymmetric OU Damping.
Test 3: Sensor Heteroscedasticity	IITM Suburban vs. BMC Urban/Industrial Variance	Levene Test: **$F = 75.75$, $p = 2.66 \times 10^{-48}$**	Proved severe variance stratification between IITM (Borivali, Malad) and BMC (Byculla, Shivaji Nagar). Enabled station-specific loss regularizations.
Test 4: CAMS Satellite Chemistry F-Tests	Independent Predictive Power of Satellite Layers	Partial F-Tests: `cams_pm25_roll24` F = 3200.1, `cams_co` F = 1143.3 (all **$p < 10^{-100}$**)	11 Copernicus CAMS atmospheric chemistry and AOD variables validated cross-gap feature bridging across the 31-hour forecast horizon.
Test 5: Out-of-Sample Private-LB Verification	16h–63h Multi-Window Private LB Generalization	Paired Diebold-Mariano: **DM = +2.43, p = 0.0153**	Proved Slot #6 Dual-Regime Statistical Champion significantly outperforms Slot #5 across all 4 unseen November holdout windows.

> **Key Algorithmic Breakthrough: Dual-Regime Forecasting**
>
> Standard gradient boosters catastrophic overshoot when evaluating post-firecracker relaxation. By combining the **Asymmetric Thresholded Ornstein-Uhlenbeck (OU) Decay** with light-gradient-boosting models, our system reduces error by **54.26%** while preserving an ultra-tight **8.6979 µg/m³** accuracy across baseline winter smog.

PILLAR 3 Real-World Civic & Public Health Impact for Mumbai

Mumbai AirPulse 2026 converts raw meteorological data into tangible, actionable civic interventions across Greater Mumbai:

🚴 Citizens & Commuters

Pinpoints the **Safe Outdoor Window (14:00–17:00 IST)** when sea breezes ventilate the city (>2,400 m²/s) vs. **Nocturnal Inversion Trap Hours (01:00–08:00 IST)** when the boundary layer drops to 120m, trapping toxic smoke at street level.

🚛 BMC Ward Officers

Delivers ward-level **GRAP Stage-II/III misting tanker dispatch** schedules, construction dust curfew notices when ventilation drops below 1,200 m²/s, and heavy-goods vehicle diversion windows along the Eastern Express Highway.

🏥 Hospitals & Schools

Provides **48-hour forward alerts** to KEM and Sion hospitals to pre-stage pediatric nebulizers and asthma triage beds, and advises school principals across 24 wards to shift morning physical education indoors.

PILLAR 4 Full-Stack Functionality & Interactive Walkthrough

The production deployment at https://vercelmumbaiairpulse.vercel.app/ (https://vercelmumbaiairpulse.vercel.app/) is fully operational with zero latency and high interactivity:

🗺️ 32-Station Map Twin

Interactive Carto Dark-Matter digital twin with color-coded CPCB pins, station hover telemetry, and 48-hour timeline scrubber from Nov 05 through Nov 12.

🎛️ What-If Physics Simulator

Interactive sliders for Boundary Layer Height (50–2500m), Wind Speed (0.2–12 m/s), RH (20–95%), and CAMS AOD to simulate real-time dispersion and hygroscopic growth.

📈 Dual-Scale Trajectory Charts

Synchronized dual-axis Plotly charts with instant toggle between **Normal Smog Zoom (0–220 µg/m³)** and **Full Firecracker Scale (0–820 µg/m³)**.

Edge Architecture & Neon Postgres: Powered by serverless endpoints (`/api/telemetry` and `/api/alerts`) connected to a Neon Serverless PostgreSQL instance with automatic bundled edge-JSON failover, ensuring instant response times and 100% reliability.

Submission: Google Mumbai Air Quality Forecasting Challenge ($500 Creative Track)

Author: Hitanshu Chachad (`hitanshuac@gmail.com`)

Live: vercelmumbaiairpulse.vercel.app